Solving the AC Power Flow Problem with Newton-Raphson

Theory, Implementation, and Empirical Sensitivity Analysis on a 7-Bus System

Abstract

The AC power flow problem, which is used to determine the voltage magnitude and phase angle at every node of an electrical network given its specific power injections, is the computational foundation of power systems engineering. It is nonlinear, has no closed-form solution beyond trivial cases, and must be solved millions of times per day across various of the world's grids.

This paper develops the problem from the first principles, originally derived from the Newton-Raphson formulation in polar coordinates, presents a complete Python implementation, and then subjecting it to an empirical study on a 7-bus test system. Every numerical result reported here was produced by the accompanying code.

The empirical findings consist of:

- The solver converging in 3-4 iterations to a tolerance of $10^{-8} p.u.$, with the direct mismatch ratio of $\frac{m_k}{m_{k-1}^2}$ which settles at a near-constant that equals about 0.085, a direct and measured confirmation of quadratic convergence.
- An N-1 contingency screen of all eight branches identifies two critical lines (Line 1-4 and Line 6-7) whose loss drives a bus outside the $\pm$ 5% voltage band.
- A load-continuation study locates the system's maximum load at $\lambda \approx 5.151$ which equates to 1082MW or a 415% margin over base load. At that point the Newton-Raphson Jacobian becomes numerically singular, with its smallest singular value falling from 4.678 at base case to 0.011, and its condition number explodes from 10.1 to 2669. Voltage collapse is a physical blackout mechanism that is shown to be mathematically identical to the Jacobian losing rank.

The final result is the core of this paper, with the numerical method of said failure not being a bug it is the physical law.

What is the Problem Being Solved

1.1 - Physical Setting

Electrical power grids are a network of buses defined as substations, or more abstractly, electrical nodes connected through transmission lines and transformers, also known as branches. Generators inject power at some buses, with loads withdrawing it at others, while the branches carry the difference.

A natural but incorrect model is that but incorrect mental model is that operators send power from a generator to a load along a chosen path. This is not the case, as power flow is not something that is routed. Power distributes itself across every available path according to the physics of the network, in proportion to the electrical admittances involved. An operator can choose how much a generator produces; resulting in voltages everywhere, and the flow on every individual line, which are then determined by Kirchoff's laws and are not independently choosable

This is why power flow analysis exists; to answer the question of:
Given the amount of power each generator is producing and each load is consuming, what voltage magnitude and phase angle results at every bus, and how much power consequently flows on every branch?

1.2 - The Difficulty of This Problem

Two facts take this problem out of the trivial sphere:

1. The relationship between power and voltage is nonlinear. Power is a product of voltage and current, and current is proportional to voltage differences. The resulting equations (derived in §3) are quadratic in voltage magnitude and trigonometric in phase angle. For anything beyond a 2-bus system, no algebraic solution exists, making it necessary to attack the problem numerically and iteratively.
2. The equations are simultaneous and global. The voltage at bus 4 depends on the voltage at bus 3, which depends on bus 2, which depends back on bus 4. Nothing can be solved in isolation; the entire network must be resolved all at the same time.

1.3 - The Importance

Power flow is the load-bearing computation of the electrical industry:

Domain	What Power Flow Does
Interconnection Studies	Whether a proposed wind farm, solar plant, or data center may connect to the grid, and who pays for any required upgrades
Operational Security (N-1)	Whether the system would survive the loss of any single line or generator, checked continuously in control rooms
Voltage Stability	How much additional load the system can absorb before collapsing. This directly implicated in real blackouts
Electricity Markets	Which generators are dispatched, and the locational marginal prices that appear on consumer bills
Distribution Planning	How much rooftop solar or EV charging a neighborhood feeder can host

An example of the importance of understanding power flow comes from the 2003 Northeast Blackout, which left roughly 55 million people without power across the USA and Canada. The result was caused largely due to a voltage-collapse and cascading-overload event, precisely the failure that §6.4 of this paper reproduces numerically on a 7-bus system.

2. Network Model

2.1 - Per-Unit Normalization

All quantities are expressed in per-unit (p.u.), with each value being divided by a chosen base. This paper uses a 100 MVA base, so a load of 0.80 p.u. means 80 MW. Voltages are normalized so that 1.0 p.u. is nominal. Per-unit is a universal power engineering unit because it allows transformers with different voltage ratings to be compared, through keeping quantities within an order of magnitude of 1.0, and makes it simple to check if a voltage is acceptable by comparing it within a range of 0.95-1.05.

2.2 - Branch (π) Model

Each transmission line is represented by a π-model, which is defined as a series impedance $z = r + jx$ carrying the flow, plus a shunt susceptance b split in half at each end representing the line's capacitance to ground.

- r (resistance) - Dissipates real power as heat, this is where loss comes from
- x (reactance) - The dominant term at transmission voltages ($x \gg r$ typically), governing how much voltage drops across the line.
- b (charging) - Injects reactive power to mildly support the voltage.

Converting to admittance, leaves the equation: $y_s = \frac{1}{z} = \frac{1}{(r+jx)}$

2.3 - Bus Classification

Each bus has four associated quantities:

- Voltage magnitude $|V|$
- Phase angle θ
- Real Power P
- Reactive Power Q

Exactly two are known and two are unknown at each bus. Which variables are known is determined by what is physically attached:

Type	Known	Unknown
Slack (Exactly one)	$\|V\|$, $\theta = 0$	$P,\ Q$

PV (Generator	$P,\ \lvert V\rvert$	$Q,\ \theta$
PQ (Load)	$P,\ Q$	$\lvert V\rvert,\ \theta$

2.3.1 - Physical Justification of Buses

- Slack buses:
    1. The angle references: because only angle differences are physically meaningful, so one bus must be fixed at $\theta = 0$.
    2. Power Balance: Losses are unknown until the finalization of the equation , so total generation cannot be specified in advance. One bus must be left free to supply whatever the rest of the system fails to account for, taking up the “Slack” of the system.
- PV (Generator):
  The generator is dispatched to a real-power setpoint, and its automatic voltage regulator holds terminal voltage at a target by adjusting the excitation. The reactive power it must produce to achieve that voltage is an output, rather than an input.

2.3.2 - Degrees of Freedom Check

For n buses with n_{PV} generator buses:

- Unknown angles: $n - 1$ for every bus except the slack bus
- Unknown magnitudes: $n - 1 - n_{PV}$ for PQ buses only
- Total unknowns: $2n - 2 - n_{PV}$

The total equations available:

- Real-power equations at PV and PQ buses: n-1
- Reactive-power equations at PQ buses only: $n - 1 - n_{PV}$
- Total equations: $2n - 2 - n_{PV}$ ✓

The system acts as a square. This exact bookkeeping determines the dimensions of the Jacobian in §3.3 and is why the implementation constructs index lists pv_pq and pq as the first step.

3. The Mathematics

3.1 - Admittance Matrix (Ybus)

Applying Kirchoff’s current law at every bus gives $I = Y_{Bus}V$, where the bus admittance matrix is assembled methodically through each branch. Write $Y_{Bus} = G + jB$ to measure the conductance and susceptance. The Y_{Bus} completely encodes the network’s topology and impedances, is symmetric with absent phase-shifting transformer, and is highly sparse in real systems, with a 10,000-bus grid only having around 3 nonzeros per row, something production solvers exploit heavily.

3.2 - Power Flow Equations

Complex power injected at bus i is $S_i = V_i I_i^*$, and $I_i = \sum_k Y_{ik} V_k$. Substituting writing $V_i = |V_i| e^{j\theta_i}$, and separating real and imaginary parts yields the two equations that define this power systems analysis:

$$P_i = \sum_{k=1}^{n} |V_i||V_k|(G_{ik} cos\theta_{ik} + B_{ik} sin\theta_{ik})$$

$$Q_i = \sum_{k=1}^{n} |V_i||V_k|(G_{ik} sin\theta_{ik} - B_{ik} cos\theta_{ik})$$

Where $\theta_{ik} = \theta_i - \theta_k$.

3.3 - Newton-Raphson

Newton's method solves f(x) = 0 through repeated linearization. Taking a first-order Taylor expansion about the current iterate and setting it to zero:

$$f(x^k) + J(x^k)\Delta x = 0 \rightarrow J\Delta x =- f(x^k),\ x^{(k+1)} = x^k + \Delta x$$

For power flow, it is necessary to define the state vector as well as the mismatch function:

$$\mathbf{x} = \begin{bmatrix} \boldsymbol{\theta}_{PV \cup PQ} \\ \mathbf{V}_{PQ} \end{bmatrix}, \qquad \mathbf{f}(\mathbf{x}) = \begin{bmatrix} \Delta \mathbf{P} \\ \Delta \mathbf{Q} \end{bmatrix} = \begin{bmatrix} \mathbf{P}^{spec} - \mathbf{P}^{calc}(\mathbf{x}) \\ \mathbf{Q}^{spec} - \mathbf{Q}^{calc}(\mathbf{x}) \end{bmatrix}$$

The mismatch is defined as the discrepancy between the power that is injected into each bus and the power the current voltage guess actually implies. Driving it to zero solves the problem.

The Jacobian is partitioned into four blocks:

$$\begin{bmatrix} \Delta \mathbf{P} \\ \Delta \mathbf{Q} \end{bmatrix} = \underbrace{\begin{bmatrix} \partial \mathbf{P}/\partial \boldsymbol{\theta} & \partial \mathbf{P}/\partial \mathbf{V} \\ \partial \mathbf{Q}/\partial \boldsymbol{\theta} & \partial \mathbf{Q}/\partial \mathbf{V} \end{bmatrix}}_{\mathbf{J}} \begin{bmatrix} \Delta \boldsymbol{\theta} \\ \Delta \mathbf{V} \end{bmatrix} = \begin{bmatrix} \mathbf{J}_{11} & \mathbf{J}_{12} \\ \mathbf{J}_{21} & \mathbf{J}_{22} \end{bmatrix} \begin{bmatrix} \Delta \boldsymbol{\theta} \\ \Delta \mathbf{V} \end{bmatrix}$$

Differentiating the power flow equations gives the closed forms. Diagonal and off-diagonal entries differ due to the bus's sensitivity to its own state being structurally distinct from its sensitivity to a neighbor's.

$J_{11} = \frac{dP}{d\theta}$:

$$J_{11,ii} = -\ Q_i - B_{ii}V_i^2, \qquad J_{11,ik} = V_iV_k(G_{ik}sin\theta_{ik} - B_{ik}cos\theta_{ik})$$

$J_{12} = \frac{dP}{dV}$:

$$J_{12,ii} = \frac{P_i}{V_i} + G_{ii}V_i, \qquad J_{21,ik} = V_i(G_{ik}cos\theta_{ik} + B_{ik}sin\theta_{ik})$$

$J_{21} = \frac{dQ}{d\theta}$:

$$J_{21,ii} = P_i - G_{ii}V_i^2, \qquad J_{21,ik} = -\ V_iV_k(G_{ik}cos\theta_{ik} + B_{ik}sin\theta_{ik})$$

$J_{22} = \frac{dQ}{dV}$:

$$J_{22,ii} = \frac{Q_i}{V_i} - B_{ii}V_i, \qquad J_{22,ik} = V_i(G_{ik}sin\theta_{ik} - B_{ik}sin\theta_{ik})$$

J is rebuilt at every iteration, due to its dependents on the current V and θ. This is the method's main computational cost, and the reason variants like Fast-Decoupled Load Flow which freezes an approximate Jacobian exist.

3.4 - The Algorithm

1. Give a flat start by setting V = 1.0 p.u. at PQ buses and $\theta\ =\ 0$ everywhere. This is an educated guess because real grids are designed to sit near nominal voltage with small angle spreads.
2. Evaluate $P^{Calc}\ and\ Q^{calc}$ from the current state.
3. Form the mismatch of $\Delta P\ and\ \Delta Q$ over the appropriate bus subsets.
4. Test convergence: if max$|mismatch| <\ \epsilon$, stop.
5. Build J at the current state.
6. Solve the linear system $J\Delta x\ =\ mismatch$.
7. Update $\theta\ +=\ \Delta\theta\ (PV\ and\ PQ\ buses),\ V\ +=\ \Delta V\ (PQ\ buses\ only)$.
8. Repeat from step 2.

3.5 - Speed of Newton-Raphson Convergence

Newton's method exhibits quadratic convergence near a solution, meaning is e_k is the error at iteration k, then $e_{k+i} \approx Ce_k^2$. Informally, the number of correct digits doubles for each iteration.

This is what makes the Newton-Raphson method more advantageous to use over the earlier Gauss-Seidal method, which converges only linearly and can require hundreds of iterations in many typical cases. Newton-Raphson's 3-5 iterations, combined with sparse-matrix techniques for the linear solution is what made real-time power system analysis computationally feasible, and the industry standard since Tinney and Hart's 1967 formulation.

The tradeoffs that come with the Newton-Raphson are most prevalent in it requiring an initial guess, and requires the Jacobian to be nonsingular. §6.4 shows what happens when that second condition fails, making the failure informative rather than an inconvenience.

3.6 - Post-Processing: Branch Flows and Losses

Once V, and θ are known, branch flows follow from direct circuit analysis. For a branch $f \rightarrow t$:

$$I_{ft} = (y_s + j\frac{b}{2})\frac{V_f}{\tau^2} - \frac{y_s}{\tau}V_t, \qquad S_{ft} = V_f I_{ft}^*$$

- τ - The transformer tap ratio

For a plain line the transformer tap ratio is equal to one. Computing S_{ft} and S_{tf} from both ends and summing gives the branch loss $S_{loss} = S_{ft} + S_{tf}$, whose real part is the I^2R heating loss.

3.7 - Validity Check

At a true solution, the generation in total must equal total load plus total losses. This is something that is not enforced by the solver, emerging only if the equations were solved correctly. It is therefore a valid test, and the implementation passes it to machine precision (§5.2).

4. Implementation

The code is structured with the hope to mirror the mathematics one-to-one:

Mathematical Object	Code
Bus data and their type classification	class Bus
Branch π-model, y_s = 1/z	Class Branch
Y_{bus} assembly	PowerFlowNetwork.build_ybus()

Bus-type index sets	PowerFlowNetwokr._classify()
P_i, Q_i equations (§3.2)	calc_PQ()(nested in solve())
Mismatch vector	dP, dQ, mismatch
J_{11} ... J_{22} (§3.3)	J11, J12, J21, J22 → np.block()
$J\Delta x = f$	np.linalg.solve(J, mismatch)
Branch flows (§3.6)	PowerFLowNetwork.branch_flows()

4.2 - The Core Loop

The essential machinery, taken from nr_solver,py:

```
def calc_PQ(V, theta):
    P = np.zeros(n); Q = np.zeros(n)
    for i in range(n):
        for k in range(n):
            ang = theta[i] - theta[k]
            P[i] += V[i]*V[k]*(G[i,k]*np.cos(ang) + B[i,k]*np.sin(ang))
            Q[i] += V[i]*V[k]*(G[i,k]*np.sin(ang) - B[i,k]*np.cos(ang))
    return P, Q

for it in range(1, max_iter + 1):
    P_calc, Q_calc = calc_PQ(V, theta)

    dP = P_spec[pv_pq] - P_calc[pv_pq]      # mismatch at PV + PQ buses
    dQ = Q_spec[pq]    - Q_calc[pq]         # mismatch at PQ buses only
    mismatch = np.concatenate([dP, dQ])

    if np.max(np.abs(mismatch)) < tol:
        converged = True
        break

    # ... build J11, J12, J21, J22 per the closed forms of §3.3 ...
    J  = np.block([[J11, J12], [J21, J22]])
    dx = np.linalg.solve(J, mismatch)

    dtheta, dV = dx[:len(pv_pq)], dx[len(pv_pq):]
    for a, i in enumerate(pv_pq): theta[i] += dtheta[a]
    for a, i in enumerate(pq):    V[i]     += dV[a]
```

The index lists pv_pq and pq are the mechanism through which slack-bus angle and PV-bus voltage magnitude are held fixed, and they are precisely what makes the Jacobian square and stays consistent with the DOF count of §2.3.

4.3 - Test System

A 7-bus network: one slack bus, two PV generators, four PQ loads, connected by eight branches in a partially meshed topology. Impedances are in a realistic range for a transmission system of about 230 kV on a 100 MVA base:

```
net.add_bus(name="Bus 1 (Slack)", bus_type="slack", Vset=1.04)
net.add_bus(name="Bus 2 (Gen)",   bus_type="PV", Pg=0.60, Vset=1.03)
net.add_bus(name="Bus 3 (Load)",  bus_type="PQ", Pd=0.55, Qd=0.13)
net.add_bus(name="Bus 4 (Load)",  bus_type="PQ", Pd=0.80, Qd=0.30)
net.add_bus(name="Bus 5 (Load)",  bus_type="PQ", Pd=0.40, Qd=0.10)
net.add_bus(name="Bus 6 (Gen)",   bus_type="PV", Pg=0.70, Vset=1.02)
net.add_bus(name="Bus 7 (Load)",  bus_type="PQ", Pd=0.35, Qd=0.09)

net.add_branch(0, 1, r=0.010, x=0.070, b=0.020, name="L1-2")
net.add_branch(1, 2, r=0.015, x=0.090, b=0.025, name="L2-3")
net.add_branch(0, 3, r=0.012, x=0.080, b=0.022, name="L1-4")
net.add_branch(2, 3, r=0.014, x=0.085, b=0.020, name="L3-4")
net.add_branch(3, 4, r=0.018, x=0.100, b=0.028, name="L4-5")
net.add_branch(0, 5, r=0.011, x=0.075, b=0.021, name="L1-6")
net.add_branch(5, 6, r=0.016, x=0.095, b=0.024, name="L6-7")
net.add_branch(4, 6, r=0.013, x=0.082, b=0.023, name="L5-7")
```

The total load is 210 MW/62 MVAr, with a total scheduled non-slack generation of 130 MW.

5. Base Case Results

5.1 - Solution

The solver converges in 3 iterations to $10^{-8} p.u.$

Bus	Type	V (p.u.)	θ (deg)	P (MW)	Q (MVAr)
Bus 1	Slack	1.0400	0.000	82.47	76.28
Bus 2	PV	1.0300	-0.072	60.00	0.52
Bus 3	PQ	1.0072	-3.112	-55.00	-13.00
Bus 4	PQ	1.0038	-3.492	-80.00	-30.00

Bus 5	PQ	0.9953	-4.136	-40.00	-10.00
Bus 6	PV	1.0200	0.434	70.00	-18.13
Bus 7	PQ	1.0002	-2.839	-35.00	-9.00

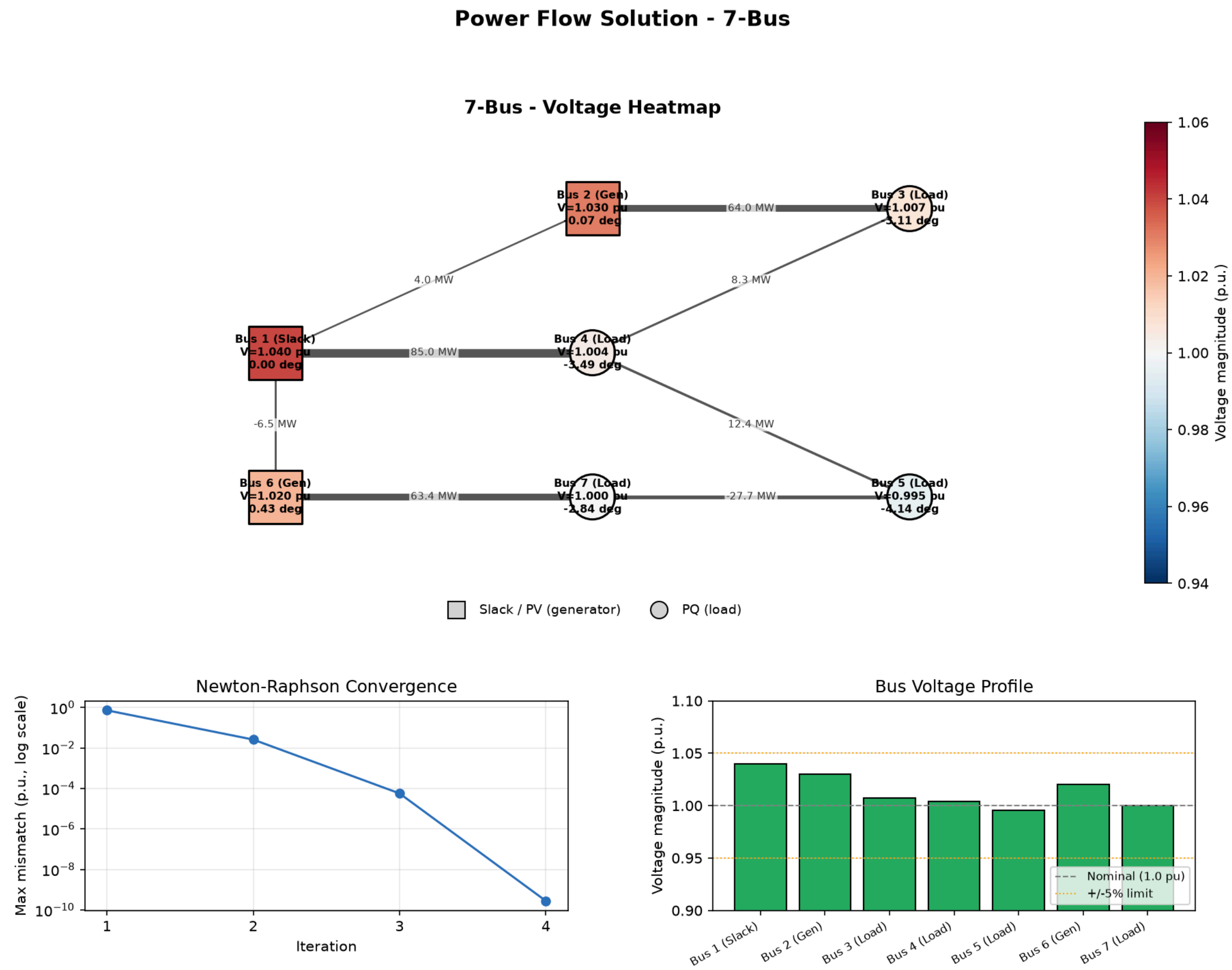

Figure 1 - Base Case Solution

The top left details the network diagram with buses colored by voltage magnitude, and the line thickness being scaled by MW flow. The bottom left graph details the mismatch per iteration on a log scale showing the convergence after the third iteration. The bottom right determines the bus voltage profile against the $\pm$ 5% band, with all the voltage lying within that percentage.

Angles at load buses are small and negative, and power flows downhill in angle, from generators toward loads, exactly as expected from the physics required of this system.

It is important to consider how the PV buses behave. Bus 2 and bus 6 hold their voltage setpoints of 1.03 and 1.02 exactly, and their reactive output of Q = 0.52 and -18.13 MVAr is a solved consequence of holding those setpoints, with bus 6 actually absorbing reactive power.

5.2 - Branch Flows and the Power Balance Check

Branch	P_from (MW)	Q_from (MVAr)	P loss (MW)
L1-2	3.977	13.209	0.020
L2-3	63.957	15.731	0.619
L1-4	84.997	35.482	0.951
L3-4	8.337	1.609	0.011
L4-5	12.373	5.009	0.035
L1-6	-6.508	27.593	0.088
L6-7	63.404	11.092	0.642
L5-7	-27.662	-2.386	0.101

5.2.1 - Power Balance

$$\sum(Generation) - \sum(Load) - \sum(Losses)$$

$$= 212.466 - 210.000 - 2.466$$
$$= 0.000000\ MW$$

Closure to machine precision. Since the solver never enforces this, it constitutes a genuine independent verification that the equations of §3.2 were solved correctly.

Two details worth noticing are the first flow on the L1-6 is negative, -6. MW specifically. This is because the generator bus 6 produces 70 MW but its neighborhood consumes less, so the surplus is flowing back toward the slack bus. The second is system losses are recorded at 2.466 MW, or 1.17% of load, which falls under the range of expected losses for a transmission network.

5.3 - Convergence

Measured mismatch history, with quadratic-convergence ratio $\frac{m_k}{m_{k-1}^2}$:

Iteration	Max Mismatch (p.u.)	$m_k \,/\, m_{k-1}^2$
1	7.2665×10^{-1}	-
2	2.6086×10^{-2}	0.049
3	5.7141×10^{-5}	0.084
4	2.7940×10^{-10}	0.086

The ratio settles to a near-constant of about 0.085, which is exactly the signature of quadratic convergence $e_{k+1} \approx Ce_k^2$ with $C \approx 0.085$. The mismatch falls nine orders of magnitude in three iterations. This is a measured confirmation of theory in §3.5, not an assertion of it.

6. Sensitivity Study

Every result below was obtained by changing one variable in the base case and re-solving, due to Newton-Raphson's observed differences being caused by a single change.

6.1 - Bus Variables

a) Set slack voltage setpoint: 1.04 → 0.98 p.u.

Bus	1	2	3	4	5	6	7
ΔV (p.u.)	-0.0600	0.0000	-0.0185	-0.0356	-0.0231	0.0000	-0.0125

Every PQ bus voltage shifts down. The PV buses 2 and 6 do not move at all, as they are pinned to their setpoints, simply absorbing the change by adjusting reactive output. This is the cleanest possible demonstration of what a “voltage-controlled bus” actually is. It is important to consider the shift is not uniform as as bus 4, which is electrically closest to slack, via the heavily-loaded line attaching bus 1 and 4, tracks the most strongly with a change in voltage of -0.0356, while bus 7, sitting near the generator bus 6, barely moves with a change of -0.013, showing that voltage support is a local phenomenon.

b) PV setpoint at bus 2: 1.03 → 1.08 p.u.

Quantity	Base	Modified
Bus 2 voltage	1.0300	1.0800 (Holds)
Bus 2 reactive output	0.52 MVAr	99.67 MVAr

Bus 6 reactive output	-18.13 MVAr	-23.25 MVAr
System losses	2.466 MW	2.862 MW

This is the most instructive experiment in the study, because it shows that raising the voltage target by a mere 5% causes a 190-fold increase in reactive output from the generator, from essentially nothing to nearly 100 MVAr. Voltage is not free. It is purchased with reactive power, and the price is steeply nonlinear. This is precisely why real generators have reactive capability limits and why enforcing them matters

c) Doubling the load bus 4 (80 → 160 MW, 30 → 60 MVAr)

Bus	**1**	**2**	**3**	**4**	**5**	**6**	**7**
ΔV (p.u.)	0.0000	0.0000	-0.0125	-0.0230	-0.0154	0.0000	-0.0088

Slack power rises from 82.47 to 165.01 MW, and losses rise from 2.466 to 5.011 MW.

Two Notes are recorded from this trial:

1. Voltage depression propagates and attenuates with electrical distance, shown by the affected bus drops the most with a change of -0.023, its neighbors less, and the distant buses the least with a change of -0.009.
2. Losses grow superlinearly, with the load at one bus rising to 80 MW (a 38% increase) but losses rose 103%. Losses scale roughly with the square of current, causing direct economic consequences, and a necessity to put urgent care into where load is sited.

6.2 - Branch Variables: the r versus x distinction

This pair of experiments isolates the two most commonly conflated parameters. Both triple a parameter on the same line, L1-4.

	Base	**Triple x(0.080→0.240)**	**Triple r (0.012→0.036)**
Bus 4 voltage	1.0038	0.9821 (-0.0217)	0.9919 (-0.0119)
Flow on L1-4	84.997 MW	48.581 MW (-43%)	87.221 MW (+2.6%)
Loss on L1-4	0.951 MW	0.333 MW	2.749 MW (x2.9)
Total System Losses	2.466 MW	3.037 MW	4.329 MW

- Reactance (x) governs flow distribution and voltage drop. Tripling it caused power to reroute away from L1-4, with flow collapsing from 85 to 49 MW. This is a result of power innately following the path of least reactance.
- Resistance (r) governs losses. Tripling it left the flow essentially unchanged but nearly tripled the lines losses as $P_{loss} = I^2R$ predicts when I barely moves.

Under tripled x, the loss on L1-4 itself falls from 0.951 MW to 0.333 MW, yet the total system sees a rise in power loss from 2.466 MW to 3.037MW. There is no contradiction in here as I^2R on that line dropped because of the current drop, but the 36 MW that fled from the line connecting bus 1 and 4 did not vanish, but instead was rerouted onto longer, higher-impedance paths, and the losses it incurred there were able to offset the local savings. This gives a final verdict of: impeding one line lowers that line's losses and raises the system's.

Under tripled r, the flow through the line connecting bus 1 and bus 2 was relatively unchanged from 85 MW to 87MW. A more resistive line carrying more power is usually seen as counterintuitive, but is possible because x >> r. If the resistance is tripled it will have little effect on the impedance magnitude, so the flow split hardly moves. At the same time the extra 1.8 MW of loss has to be supplied to the slack bus, sitting at the far end of L1-4, making the additional power needed to cover the new losses being pushed down that very line. This means the line's own inefficiency increases its loading.

To sum up, x decides where the power goes, and r decides what it costs. This is why transmission engineers speak almost exclusively in reactance when discussing the flow through the system, and why the classic DC power flow approximation neglects resistance entirely and is still able to get roughly the proper MW flows.

6.3 - Line Charging and Transformer Taps

Removing all line charging (b → 0):

Quantity	Base	b = 0
Bus 2 reactive output	0.52 MVAr	6.01 MVAr
Bus 6 reactive output	-18.13 MVAr	-12.03 MVAr
Slack reactive output	76.28 MVAr	84.00 MVAr

Voltages barely move, but the reactive burden shifts onto the generators, making the slack bus supply an extra 7.7 MVAr. Line Charging is a distributed reactive-power source that, if deleted, makes the generators make up the difference.

Adding a transformer tap ($\tau = 1.05$ on L1-2):

Quantity	Base	Tap = 1.05
All bus voltages	-	Essentially unchanged
Bus 2 reactive output	0.52 MVAr	73.29 MVAr

A result that may seem surprising at first, requiring extra time to process. Bus 2 is a PV bus that is required to hold 1.03 p.u. under all circumstances. The tap change alters the effective turns ratio, and because by definition of the PV bus it is impossible to move the voltage, the effect is instead absorbed entirely as reactive-power redistribution.

This highlights a general principle of power systems: a constraint you impose does not disappear, rather it reappears somewhere else as a cost. Had bus 2 been a PQ bus, the same tap change would have moved the voltage instead.

6.4 - N-1 Contingency Screening

Each of the eight branches was removed in turn and the system re-solved, with the standard security criterion used by every transmission operator worldwide.

Line Removed	Converged	Iters	V_min (p,u,)	Losses (MW)	Verdict
L1-2	Yes	3	0.9953	2.447	Ok
L2-3	Yes	4	0.9632	4.623	Ok
L1-4	Yes	4	0.9440	5.687	Violation (Bus 4)
L3-4	Yes	3	0.9931	2.511	Ok
L4-5	Yes	3	0.9757	2.604	Ok
L1-6	Yes	3	0.9951	2.428	Ok
L6-7	Yes	4	0.9385	5.548	Violation (Bus 7)
L5-7	Yes	3	0.9812	3.041	Ok

The system is not N-1 secure. Two of the eight branches, specifically L1-4 and L6-7 are critical, due to if either is lost it will drive a bus below 0.95 p.u.

The result is intuitive in hindsight. Both are the primary supply corridors to a heavily loaded bus, with L1-4 carrying 85 MW to bus 4 and L6-7 carrying 63 MW to bus 7. When either is lost, the power must detour through longer, higher-impedance paths, and both voltage drop and losses rise sharply, causing losses to be over double in both cases.

This is the analysis that justifies transmission investment. A planner facing this result would recommend reinforcing L1-4 and L6-7, or adding reactive support at buses 4 and 7, which is a recommendation that would have gone unnoticed if only the base case study was observed, as in the base case, every bus was within limits.

6.5 - Voltage Collapse

The most important experiment, which scales all loads to a factor λ and the system resolved at each step, tracking not just the voltages but the smallest singular value of the Jacobian, $\sigma_{min}(J)$, and its condition number.

λ	Total Load (MW)	Iters	V_min (p.u.)	Losses (MW)	σ_{min}(J)	cond(J)
1.00	210	3	0.9953	2.47	4.678	10.1
2.00	420	4	0.9488	10.96	4.308	10.4
3.00	630	4	0.8899	28.75	3.728	11.3
4.00	840	5	0.8084	61.41	2.824	13.7
4.50	945	5	0.7495	88.39	2.148	17.0
5.00	1050	6	0.6504	136.33	1.032	31.8
5.10	1071	7	0.6095	156.08	0.596	52.8
5.15	1082	9	0.5598	179.43	0.094	318.4
5.1513	1082	-	0.5511	-	0.011	2669
5.20	1092	No Convergence	-	-	-	-

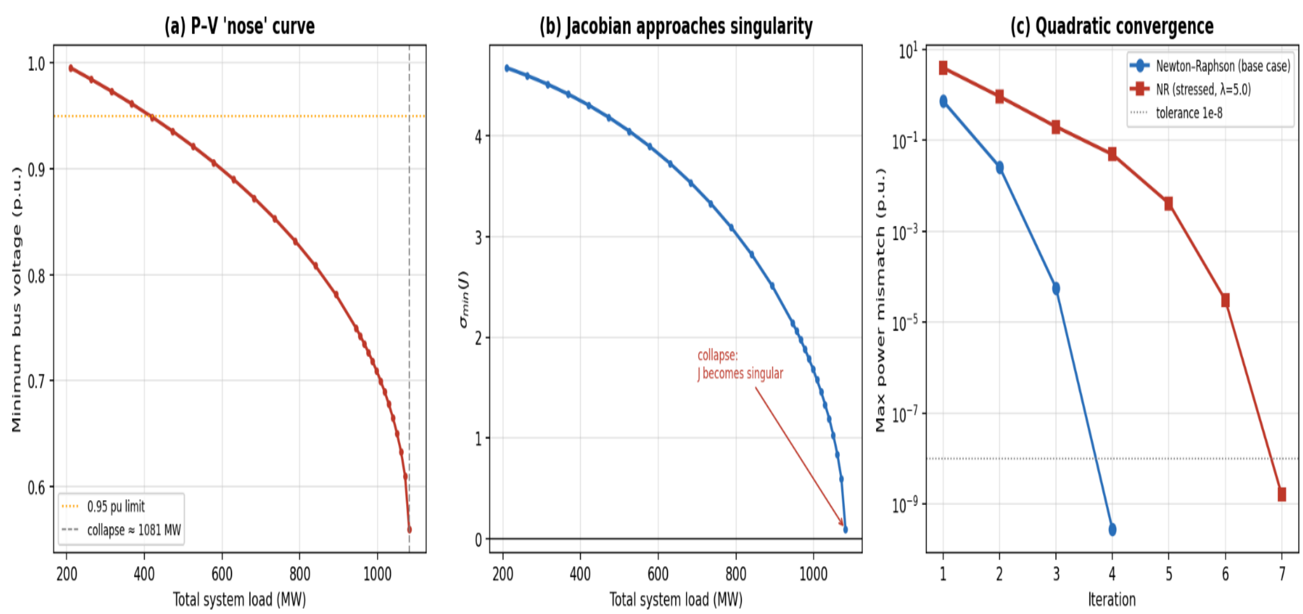

Figure 2: The P-V nose curve, the smallest singular value of the Jacobian falling to zero at the same time as the load, and the base case versus the stressed case.

Bisection locates the maximum loadability point at $\lambda^* \approx 5.151$, which results in a total load of 1082 MW, or a 415% margin above base.

Reading the three figures together tells the same story:

1. Iteration count creeps up (3→4→5→6→7→9) as the system stresses. This tells us Newton-Raphson is working harder.
2. σ_{min}(J) collapses toward zero from 4.678 to 0.011, showing the Jacobian is losing rank.
3. The condition number explodes (10→2669). The linear system $J\Delta x = f$ is becoming ill-posed.
4. Beyond λ^*, no solution exists. The solver fails, because there is simply nothing to find.

At λ^*, the Jacobian becomes singular. Mathematically, $\frac{d(P,Q)}{d(\theta,V)}$ loses invertibility, meaning that an infinitesimal increase in demand power requires an unbounded change in voltage. Physically, this is voltage collapse, defined as the point beyond which the network cannot deliver more power at any voltage, and the system falls apart.

The numerical breakdown and the physical catastrophe are the same event. The singularity of J is not an artifact of the algorithm, it is the mathematical shadow cast by a real blackout mechanism. This is why σ_{min}(J) is used in practice to act as a voltage stability check index, which is a live warning signal in energy management systems.

It is also why a power flow that fails to converge must be dismissed as a numerical nuisance, making it the potential most important result the tool gives.

6.6 - Solver Parameter

Tolerance sweep (reference: $V_4 = 1.00384764$ at $\epsilon = 10^{-8}$):

Tolerance	Iterations	Bus 4 V (p.u.)	Error vs. Reference
10^{-1}	1	1.00596029	2.1×10^{-3}
10^{-2}	2	1.00385223	4.6×10^{-6}
10^{-4}	2	1.00385223	4.6×10^{-6}
10^{-6}	3	1.00384764	0
10^{-8}	3	1.00384764	0
10^{-10}	4	1.00384764	2.2×10^{-11}

The iteration count is a step function, with tolerances of 10^{-2}, 10^{-3} *and* 10^{-4} that all terminate after two iterations; a result of quadratic convergence causing the mismatch to leap from 2.6×10^{-2} straight past all three thresholds to 5.7×10^{-5} in a single step. Quadratic convergence means paying much for a tight tolerance rarely happens, with tightening from 10^{-2} to 10^{-8} costing exactly one additional iteration.

Warm Starting:

Scenario	Iterations
Flat start, base case	3
Flat start, after +5 MW load change	3
Warm start (previous solution as initial guess), same change	2

A 33% reduction. On a 7-bus system this is a trivial problem, but when a 60000-bus system needs to be solved every few seconds, it is the difference between feasible and infeasible. This is exactly why production state estimator and real-time contingency analysis engines warm-start from the last known operating point rather than flat-starting.

7. Limitations and Caveats

7.1 - Generator Reactive Limits are Stored but not Enforced

The Bus class accepts Qmin and Qmax parameters and stores them. The solver never reads them. This is a genuine functional gap, and §6.1(b) shows exactly why it matters: raising bus 2's setpoint to 1.08 p.u. demanded 99.67 MVAr from that generator. A real machine with a 80 MVAr limit cannot do that.

The correct behavior standard in production solvers is PV→PQ bus-type switching: if a PV bus's computed Q exceeds its limit during iteration, the bus is reclassified as a PQ bus with Q fixed at the violated limit, and its voltage is allowed to fall below setpoint. The bus may switch back on later iterations. Without this, the solver silently reports operating points that no physical generator could sustain.

7.2 - Dense matrices

Ybus and the Jacobian are stored as dense NumPy arrays, and np.linalg.solve performs a dense LU factorization at $O(n^3)$. This is entirely fine at $n = 7$ and completely unusable at $n = 10000$. Real solvers exploit the extreme sparsity of Y_{bus} (a few nonzeros per row regardless of system size) using sparse LU with ordering heuristics, bringing cost to near-linear in n. Migrating to scipy.sparse would be straightforward.

7.3 - Other omissions

- **No branch thermal (MVA) ratings.** The tool reports flows but never checks whether a line is overloaded. The N−1 study in §6.4 therefore screens for *voltage* violations only — a real security assessment also screens for thermal overloads, and might well have flagged additional contingencies.
- **No phase-shifting transformers.** Tap magnitude is supported; complex tap (phase shift) is not.
- **No shunt capacitors/reactors** as bus-level devices.
- **Single slack bus, no distributed slack.** Real systems often distribute the balancing duty across multiple machines by participation factor.
- **No optimization.** This is power flow, not *optimal* power flow (§8).
- **Balanced three-phase, positive-sequence assumption.** Standard for transmission, but invalid for unbalanced distribution networks, which need a three-phase solver.

7.4 On the results themselves

The 7-bus system is synthetic. Its impedances are plausible but not drawn from a real network, and its 415% loadability margin is far more generous than a real, well-utilized grid would exhibit (real margins are more often 10–40%). The qualitative phenomena demonstrated Jacobian singularity at collapse, r/x role separation, PV-bus reactive absorption are fully general. The quantitative values are specific to this test case and should not be read as typical.

8. Extensions

Ordered roughly by effort:

1. **Enforce Q-limits via PV→PQ switching** (§7.1). Highest value per line of code.
2. **Sparse linear algebra** (§7.2). Required for any realistic system size.
3. **Branch MVA ratings + thermal violation screening.** Completes the N−1 analysis.
4. **Automated N−1 contingency engine.** The §6.4 study, but wrapped as a first-class feature with a ranked severity report.
5. **Continuation power flow (CPF).** The §6.5 study is a crude continuation. A proper CPF uses a predictor–corrector scheme with an augmented, non-singular system that can trace around the nose of the P–V curve and onto the lower (unstable) branch where plain Newton–Raphson cannot go, precisely because J is singular there.
6. **Fast-Decoupled Load Flow (FDLF).** Exploits $x \gg r$ to freeze and decouple the Jacobian into two constant matrices, factorized once. Converges linearly (more iterations) but each iteration is far cheaper, often a net win, and historically important.
7. **DC power flow.** The linear approximation: ignore r, assume $V \equiv 1$, small angles. Reduce everything to $P = B\theta$ a single linear solution. Wildly inaccurate for voltage, surprisingly decent for MW flows, and the workhorse of electricity market clearing for exactly that reason.
8. **Optimal Power Flow (OPF).** Wrap an optimizer around the power flow equations: minimize generation cost subject to the §3.2 equations as equality constraints, plus limits as

inequalities. The Lagrange multipliers on the nodal balance constraints are the local marginal prices that set wholesale electricity prices. This is a substantial step up in complexity and nonlinear constrained optimization but it is where power flow meets economics.

9. Conclusion

The AC power flow problem is small to state and deep to solve. Two equations per bus, a nonlinearity that forecloses closed-form solution, and an iterative method that converges so fast it feels like cheating nine orders of magnitude in three iterations, as measured in §5.3.

The implementation presented here is faithful to the mathematics: build_ybus() is the admittance assembly, calc_PQ() is the power flow equations, and J11–J22 are the analytic partial derivatives. It solves a 7-bus system correctly, verified independently by a power-balance closure to machine precision.

But the most valuable thing this tool produces is not a solution. It is a failure. When the load scaling reaches $\lambda^* \approx 5.151$ and Newton–Raphson stops converging, the tool is not malfunctioning it is reporting, in the only language it has, that the network has run out of physics. The Jacobian's smallest singular value falling from 4.678 to 0.011 is voltage collapse, written in linear algebra.

That correspondence between a matrix losing rank and a grid falling over is the reason this small piece of code is worth understanding properly. A tool that only tells you the answer teaches you a number. A tool that tells you where the answers stop existing teaches you the system.

Appendix A: Notation

Symbol	**Meaning**
V_i, θ_i	Voltage magnitude (p.u.) and phase angle (rad) at bus i
P_i, Q_i	Net real (MW) and reactive (MVAr) power injection at bus i
$Y_{bus} = G + jB$	Bus admittance matrix; conductance and susceptance
$y_s = 1/(r + jx)$	Series admittance of a branch
b	Total line charging susceptance (split b/2 per end)
τ	Transformer off-nominal tap ratio
θ_{ik}	$\theta_i - \theta_k$

J	Power flow Jacobian blocks J_{11} ... J_{22}
ϵ	Convergence tolerance on max power mismatch
λ	Loading scaling factor; λ^* = Maximum loadability
$\theta_{min}(J)$	Smallest singular value of the Jacobian

Appendix C: Further Reading

- Tinney, W. F. & Hart, C. E., *Power Flow Solution by Newton's Method*, IEEE Trans. PAS, 1967 — the original formulation.
- Stott, B. & Alsaç, O., *Fast Decoupled Load Flow*, IEEE Trans. PAS, 1974 — the decoupled variant.
- Grainger, J. & Stevenson, W., *Power System Analysis* — the standard textbook treatment.
- Kundur, P., *Power System Stability and Control* — the authoritative reference on voltage stability and collapse.
- Ajjarapu, V. & Christy, C., *The Continuation Power Flow*, IEEE Trans. Power Systems, 1992 — tracing around the nose curve.
- U.S.–Canada Power System Outage Task Force, *Final Report on the August 14, 2003 Blackout*, 2004.